

SWARM SATELLITE BASED GNSS IONOSPHERIC DATA ANALYSIS FOR EARTHQUAKES PREDICTION

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SWARM SATELLITE BASED GNSS IONOSPHERIC DATA ANALYSIS FOR EARTHQUAKES PREDICTION

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Institute of Space Technology
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Muhammad Sibtain Satti

ABSTRACT

By correlating the lithospheric variation in ionosphere over the epicenter of future earthquakes (EQ), satellites measuring ionospheric variation are able to corroborate the Lithosphere-Atmosphere-Ionosphere coupling (LAIC) hypothesis. In this research searches for abnormal ionospheric signatures ahead of two large EQs having magnitudes ($M_w \geq 6.5$) and shallow depths ($< 12\text{km}$) in Plasma Density (Ne), Slant Total Electron Content (STEC), and Electron Temperature (ET), which are all recorded by ESA's Swarm satellites (Swarm B, -C, and -A). An investigation of these anomalies is conducted in the data for 60 days before and after each major shock. A statistical analysis of median (M) and standard deviation (σ) is used to detect anomalies within the Dobrovolsky's area. Ionospheric data collected under quiet geomagnetic conditions have been selected for analysis, in order to exclude geomagnetic perturbations. It is clear that there is a significant perturbation in the STEC of the Swarm-B data before $M_w 6.5$ in Swarm-A, and the same is true in Swarm-B, where anomalous variations occur in ET and STEC. As for Swarm-C, no clear anomalies were associated with the USA incident. In addition, $M_w 6.7$ Turkey EQ has abnormal post-seismic Ne, STEC, and ET responses in the form of intensive perturbations exceeding the upper bound in Swarm-A, -B, and C. Aside from the Swarm observations, we also analyze the atmospheric data over the epicenters, including the Outgoing Longwave Radiation (OLR), air temperature (AT), surface temperature (ST) and relative humidity (RH). A pre-seismic atmospheric anomaly is observed by both RH and ST before the USA earthquake event, at the same anomalous time as Swarm satellites. Post-seismic atmospheric anomalies have been observed in OLR, ST and AT after Turkey EQ within 15-20 days. Highlighting the understanding of the precursor chain from the lithosphere to the atmosphere, followed by the ionosphere. It aims to determine the quality of Swarm satellites for seismo-ionospheric irregularities, which requires

further studies in order to delineate the LAIC hypothesis for future EQs. The window of variation, for both Swarm satellites (ionospheric) and atmospheric indices is five days before the USA main and one month after Turkey EQ.

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1. CHAPTER 1: INTRODUCTION

Earthquakes result from tectonic stress changes that occur inside the Earth at different hypocentral depths over long periods of time. A sudden movement of the Earth known as an earthquake result from the release of strain that has built up over a long period of time. The EQ process is triggered by tectonic forces inside the Earth during the EQ preparation period. Earth crust is under an extremely high level of stress because of this stress, which effectively causes rock ruptures on a large scale[1]. Throughout the Earth's history, plate tectonics has continually shaped it as huge plates are moved over, under, sometimes past each other by the forces of plate tectonics.

Sometimes, earthquakes are the main cause of tsunamis which may lead to the loss of life and destruction of property. With the advancement in the field of science, scientist for decades have studied this naturally occurring phenomenon and have tried to predict the earthquake using different instruments and techniques.

1.1 HISTORY

In the last decades of the 20th century, scientists began searching for EQ precursors in the atmosphere and ionosphere. First, it was put to use in structural geological and geomorphological studies. On the basis of satellite images, faults and structures were mapped. In time series analysis, this approach is quite constrained. There was no way to measure short-term processes prior to and after the earthquake. With the advancement in the field of space science, many detailed analyses for EQ precursors were done with multiple satellites data[2-3]. The present state of remote sensing application for earthquake study suggests a few earthquake-related phenomena, namely the Earth's

surface deformation, surface temperature and humidity, atmospheric temperature and humidity, gas and aerosol content.

According to the scientists, during the preparation phase of EQ different changes occur at lithosphere as well as atmosphere level. The alterations that occurred throughout the EQ preparation phase were evaluated using the Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) hypothesis using various satellite data.

1.2 Using GNSS for the Prediction of EQ

GNSS (Global Navigation Satellite Systems) consists of a constellation of satellites from GPS, GLONASS, BeiDou and Regional Navigation Satellite Systems (RNSS) such as Indian Regional Navigation Satellite System) IRNSS and QZSS Japan). Seismology today employs a wide range of satellite tools and procedures. Satellite data was originally used for earthquake investigation in the 1970s, when active faults were traced on satellite photos. In order to anticipate the Earthquake, relatively novel uses of remote sensing for seismology such as air sounding, gas measurements, and cloud analysis are being investigated.

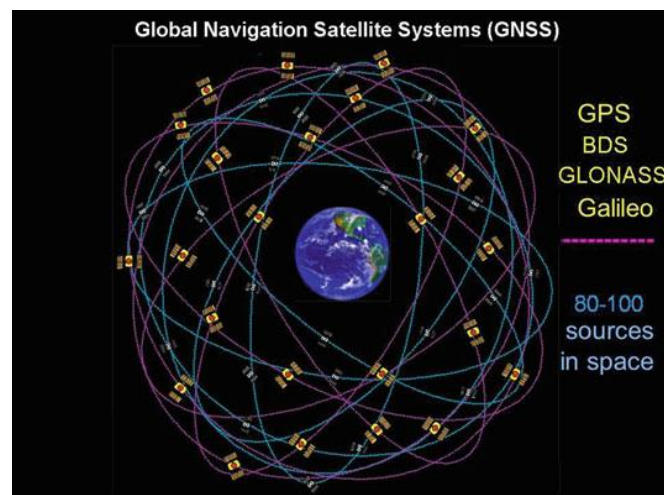


Figure 1.1: GNSS constellations

1.3 Lithosphere-Atmosphere-Ionosphere Coupling (LAIC)

To explain the LAIC effects in the atmosphere and ionosphere, various hypotheses are proposed in the form of Positive holes (p-hole) and other gas emission (e.g., radon etc.). Freund et al. postulated that strained rocks produce p-holes during the EQ preparation stage, which create electric currents and radiate electromagnetic radiation as shown in Figure B. These p-holes move upward, causing ionosphere instabilities and deviating radio signals from their normal path [4].

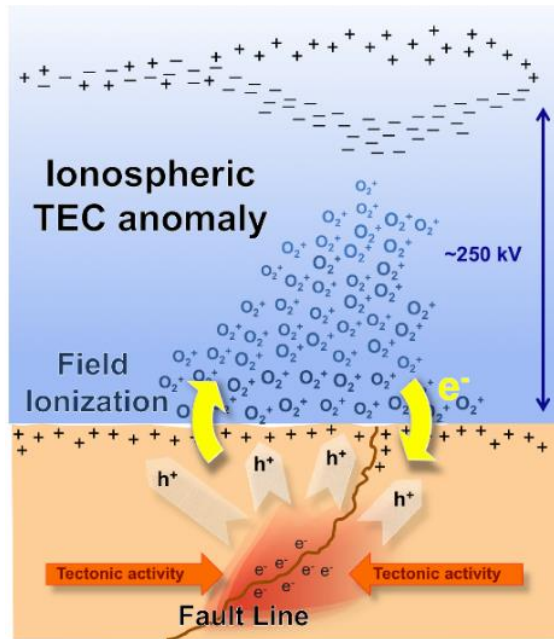


Figure 1.2: A positive airborne ion-laden air bubble expands upward through the atmosphere column, dragging along the Earth's ground potential and eliciting a polarization response in the ionosphere, resulting in electron redistribution at its lower edge.

Similarly, Pulinets et al. [5] explain the LAIC model by introducing the concept of radon emission and condensation of water. Air ionization at atmosphere and Earth owing to continual emission of radon in EQ-prone locations may ascend farther to the ionosphere, culminating in ionospheric precursors [5]. However, numerical simulation of the LAIC coupling for various EQs is theoretically presented to offer insights into this phenomenon [6-7].

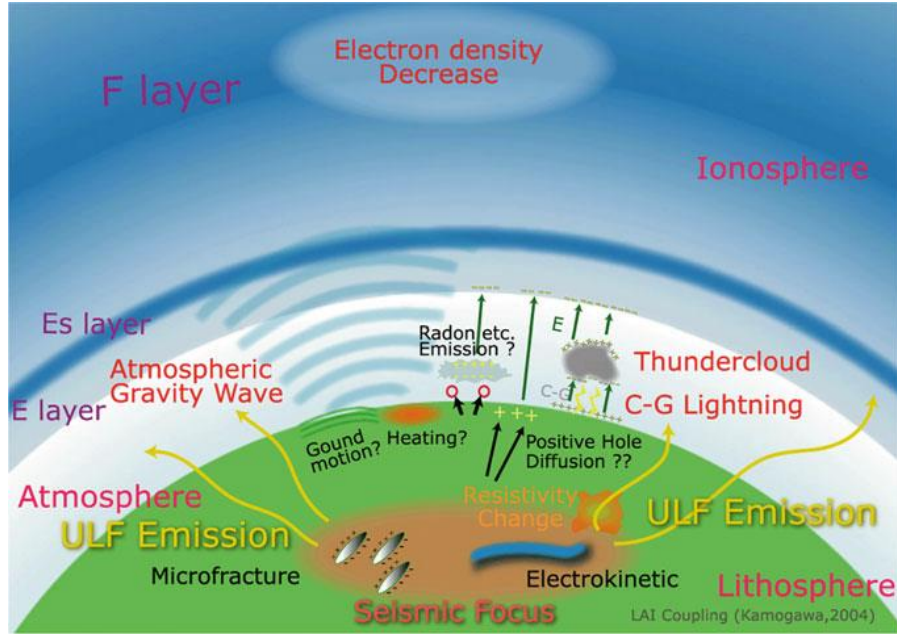


Figure 1.3: Lithospheric–atmospheric–ionospheric disturbances and coupling (Kamogawa 2006)

1.4 Ionospheric Anomalies

According to the LAIC hypothesis coupling results in the disturbance in the parameters of all these three layers. Total Electron Content (TEC), electron temperature (ET) and Electron density (N_e), being some of them which are the parameters of Ionosphere.

1.4.1 Total Electron Content (TEC)

Total electron content (TEC) is the number of free electrons along a one-meter-squared ray path that extends from the ground to the ionosphere. It is measured in TECU. 1 TECU is 10^6 electrons per square meter. The TEC is path dependent.

1.4.1.1 Slant TEC

Slant TEC (STEC) is calculated by integrating electron density along any straight line, and is calculated by the following equation

$$\begin{aligned}
STEC &= \frac{f_1^2 f_2^2}{40.28(f_1^2 - f_2^2)} (L_1 - L_2 + \lambda_1(N_1 - N_2) - \lambda_1(N_1 - N_2) + \epsilon) \\
STEC &= \frac{f_1^2 f_2^2}{40.28(f_1^2 - f_2^2)} (P_1 - P_2 - (d_1 - d_2) + \epsilon)
\end{aligned} \tag{1}$$

In the preceding equations, f_1 and f_2 are the GNSS signal's carrier phase frequencies. P and L are the pseudo-range and carrier phase observations of the GNSS signal's delay path, respectively. λ and d denote GNSS signal's wavelength and pseudo-range respectively, the signal's ambiguity is N , and b is the instrumental bias. In the GNSS signal random residual along the ray path is ϵ .

1.4.1.2 Vertical TEC (VTEC)

The vertical TEC (VTEC) is calculated by integrating electron density along a perpendicular path to the ground. The following function converts STEC to vertical TEC (VTEC)[8]

$$VTEC = STEC \cdot \cos \left(\arcsin \left(\frac{R \cdot \sin Z}{R + H} \right) \right) \tag{2}$$

Where Z is the satellite elevation angle for precise point positioning, and R and H are the radius of the Earth and the height of the top layer of the ionosphere, respectively [9-10].

1.4.2 Electron Density (N_e)

The ionosphere is the weakly ionized portion of the earth's atmosphere that extends from 50 km to over 1000 km. The main source of ionization is the sun. Ionization results in the creation of electrons which are thought to be disturbed as a result of lithosphere atmosphere ionosphere coupling within an EQ preparation period.

1.4.3 Electron Temperature (ET)

During the preparatory phase of EQ, the mechanical stress inside the rock column causes the perturbation on the Earth surface, and is eventually propagated to atmosphere and ionosphere. As a result of the propagation of these perturbations the Electron temperature (ET) changes in the ionosphere.

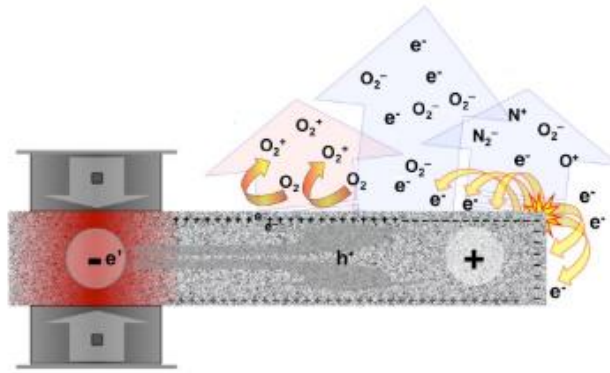


Figure 1.4: An illustration of (i) the production of positive airborne ions when the rock surface potential is positive, and (ii) the occurrence of corona discharges that produce free electrons as well as positive and negative airborne ions as the rock surface potential changes.

1.5 Study of EQ induced ionospheric anomalies using GNSS and other satellites

Many studies have been conducted on EQ-induced ionospheric anomalies on a global and regional scale, using various methods for monitoring the epicentral regions. The analysis of TEC from the Global Navigation Satellite System (GNSS) has played an important role in determining variations in the ionosphere from various ground stations and space observations [11–14]. However, there is no unbreakable link between EQ and the ionosphere (i.e., LAIC). Despite these differences, TEC analysis is quite useful in predicting the impending main EQ alignment zone, which requires additional research [15–18]. Long-term DEMETER (Detection of Electro-Magnetic Emissions Transmitted from Earthquake Regions) satellite data anomalies in ion composition were investigated by Parrot et al. [19]. He also did the same for N_e and ion temperature

in seismic region. Similarly, Shi et al. [20].investigated the 2015 M_w 7.8 Nepal EQ and associated ionospheric perturbation by using GPS for LAIC, further reported both positive and negative anomalies during EQ breed days.

Swarm satellites have recently provided an opportunity to study electromagnetic anomalies in the ionosphere to support GNSS observation. Clear concentrations of electron density and magnetic anomalies starting from almost two months before the main shock till some days before the main shock were found by analyzing the long-term electron density and magnetic field data of more than 4 years measurements of the Swarm constellation [21], [22]. A chain of quasisynchronous anomalies was also discovered by Marchetti et al. They proposed a global viewpoint in which the earthquake preparation phase affects Earth system equilibrium, resulting in anomalies in the lithosphere, atmosphere, and ionosphere for about a year [23].

1.6 Swarm-three Constellation

A dedicated mission, Swarm for the survey of Earth's magnetic field, it's temporal evolution and electric field in the atmosphere, was launched by ESA in November 2013 in near-polar orbit. This Alpha and Charlie are almost parallel to each other at a lower orbit of about 462 km above the Earth's surface, with a separation of about 1.4 degrees. The Bravo satellite, on the other hand, is in a slightly higher orbit of around 511 km. These orbital configurations were chosen primarily to measure variations in the Earth's magnetic field [24].



Figure 1.5: Swarm satellites vertically positioned, ready to attach to launch adapter. The constellation is being prepared for launch in Plesetsk, Russia. (Courtesy ESA)



Figure 1.6: The Swarm constellation's three satellites are identical. Each is 9.1 m long (including the boom), 1.5 m wide, and 0.85 m high, weighing in at 473 kg. (Courtesy ESA)

During the Swarm mission, we will gain new insights into the Earth system by improving our understanding of the Earth's interior, the Geospace environment, including the Sun-Earth connection processes. This mission will provide the best ever survey of the geomagnetic field and its temporal evolution [21]. The primary research objectives of Swarm mission are:

- Core dynamics, geo-dynamo processes, and interaction of core and mantle
- Magnetization of Lithosphere and its geological interpretation,
- 3-D electrical conductivity of the mantle,
- Flowing of currents in Ionosphere and magnetosphere.

Swarm satellites are outfitted with sophisticated magnetometers and a variety of measuring instruments to measure the Earth's magnetic field, electron density (N_e), electron temperature (ET), spacecraft potential, precise satellite orientation and attitude in space. The most important of

these instruments are the Vector Field Magnetometer (VFM) and Absolute Scalar Magnetometer (ASM), which provide precise magnitude and direction of magnetic field measurements and scalar magnetic field measurements to calibrate the onboard VFM [24].

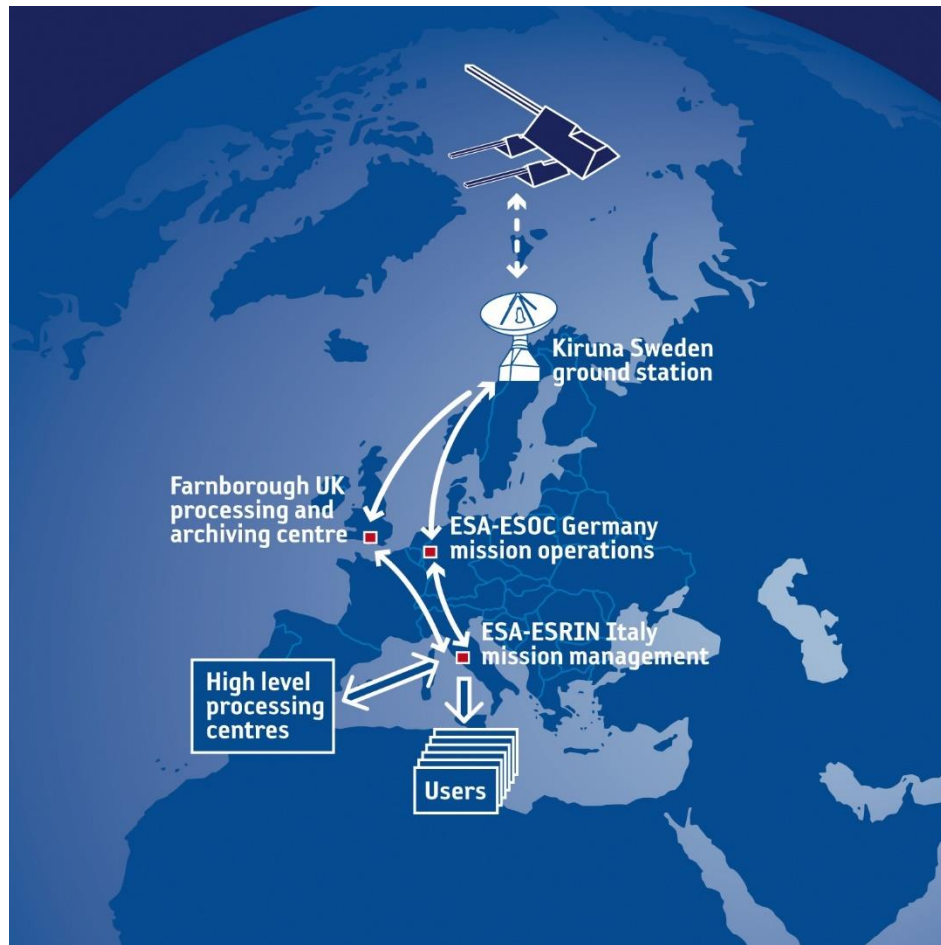


Figure 1.7: Swarm mission data flow and ground segment. (Courtesy ESA)

2. CHAPTER 2: METHOD

The ionospheric anomalies for the main shocks of two EQs are investigated by utilizing multiple sensors Swarm satellite data. Two EQs are: The M_w 6.5 USA EQ on March 31, 2020 with geographical coordinates (44.465°N, 115.118°W) and the M_w 6.7 Turkey with the location of Lat 38.431°N, Long 39.061°E, that hits the region on Jan 24, 2020 (Fig.1).

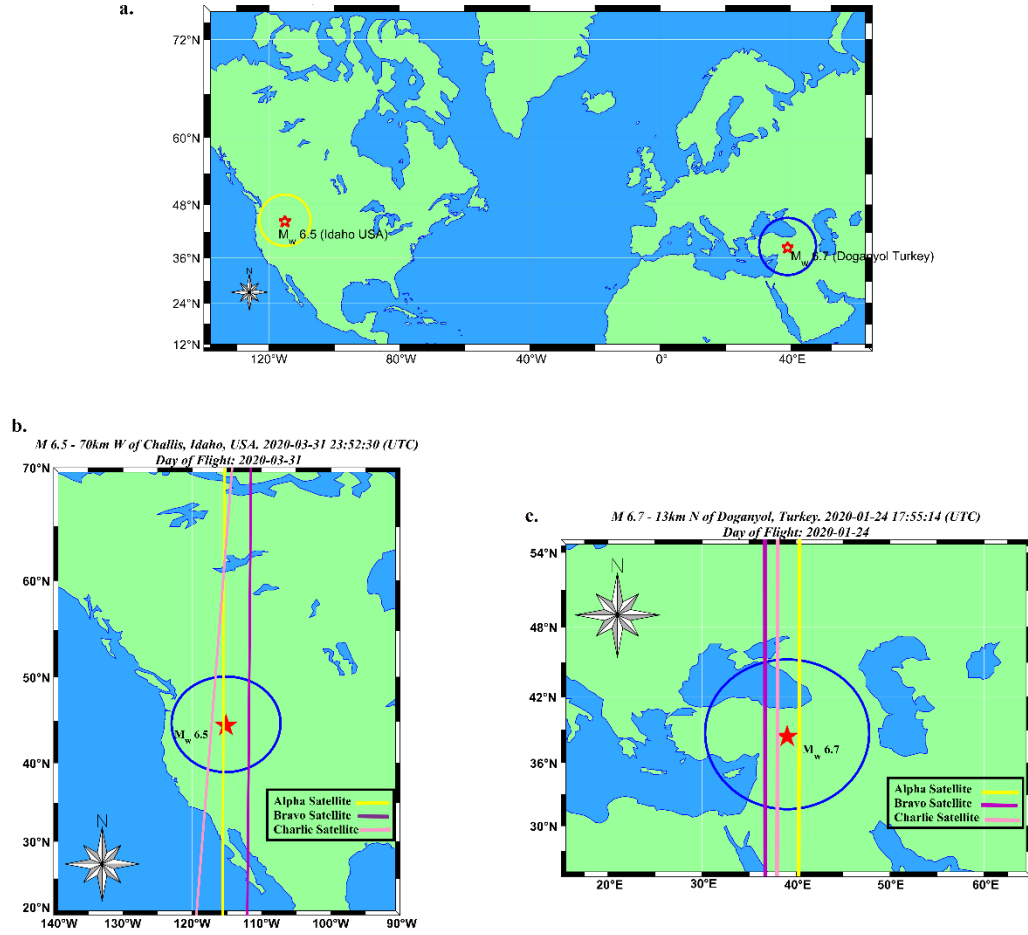


Figure 2.1: **a.** Locations of the two EQs (USA and Turkey events), which are represented by red stars. Circle's represent preparation zone estimated by Dobrovolsky's equation (Table 1). **b.** Geographical location of USA event along with the flight path of Swarm satellites on the day of EQ. **c.** Geographical location of USA event along with the flight path of Swarm satellites on the day of EQ.

The EQs information is retrieved from USGS <https://earthquake.usgs.gov/earthquakes/eventpage/us70008jr5/executive> and <https://earthquake.usgs.gov/earthquakes/eventpage/us60007ewc/executive>. The detail of geographical locations and other information are presented in Table 1.

Table 1: Information about the EQs

Sr. No	Event	Date	Latitude (°)	Longitude (°)	depth km	EQ preparation zone km	Pre-seismic Anomalous Days window		Post-seismic Anomalous Days window	
							Ionosphere	Atmosphere	Ionosphere	Atmosphere
1	6.5 (USA)	2020-03-31	44.4°N	115.1°W	12.1	623.7	0-2	0-5	Nil	Nil
2	6.7 (Turkey)	2020-01-24	38.4°N	39.0°E	10.0	760.3	Nil	Nil	20-30	15-20

The data for above mentioned EQs was studied during quite geomagnetic conditions ($K_p < 3$). Geomagnetic indices of geomagnetic storms (K_p and Dst) are studied to analyze anomalies in STEC, ET and N_e before and after the EQ's main shock (Fig. 2a,2b,2c) and (Fig. 3a,3b,3c). These geomagnetic indices are obtained from NASA's OMNI web via an online portal (<https://omniweb.gsfc.nasa.gov/form/dx1.html>).

Ionospheric data such as N_e , STEC and ET are obtained from Swarm three satellites by the online virtual service provided by ESA called VirES (The virtual research service). It provides highly interactive data manipulation for Swarm data sets. Furthermore, subsets of Swarm data selected using flexible filtering methods can be downloaded in a variety of encoding formats. Data for every EQ was downloaded by selecting the area in the VirES (<https://vires.services/>). The area

was chosen in accordance with the epicentral critical region. [25]. Data was collected in the comma separated values (CSV) format and it contained the N_e , STEC and ET for each EQ preparation area. The average of data of each satellite (Satellite Alpha, Bravo and Charlie) was downloaded separately for each EQ within the radius of the critical zone.

In this research atmospheric indices such as AT, ST, RH and OLR over the two EQs for the same time period as ionospheric parameters are also studied. These atmospheric indices' daily values for day and night can be found at the following address: <https://giovanni.sci.gsfc.nasa.gov/>. To study the LAIC's propagation over the epicentral zone, atmospheric values between the lithosphere and ionosphere were analyzed.

Swarm satellite data is analyzed in the EQ preparation zone of every event, which is estimated around the epicenter by using Dobrovolsky formula[25]. For any EQ, the stress radius is given by the following formula.

$$\rho = 10^{0.43 M} km \quad (3)$$

where M is for EQ magnitude and ρ is the critical zone of anomalous region around epicenter. The radius of the USA EQ is 623.7 km whereas radius of Turkey EQ is 760.3km. Furthermore, the magnitude of future EQ influences the radius of the critical region; for example, a large magnitude has a large preparation zone, and vice versa. [3].

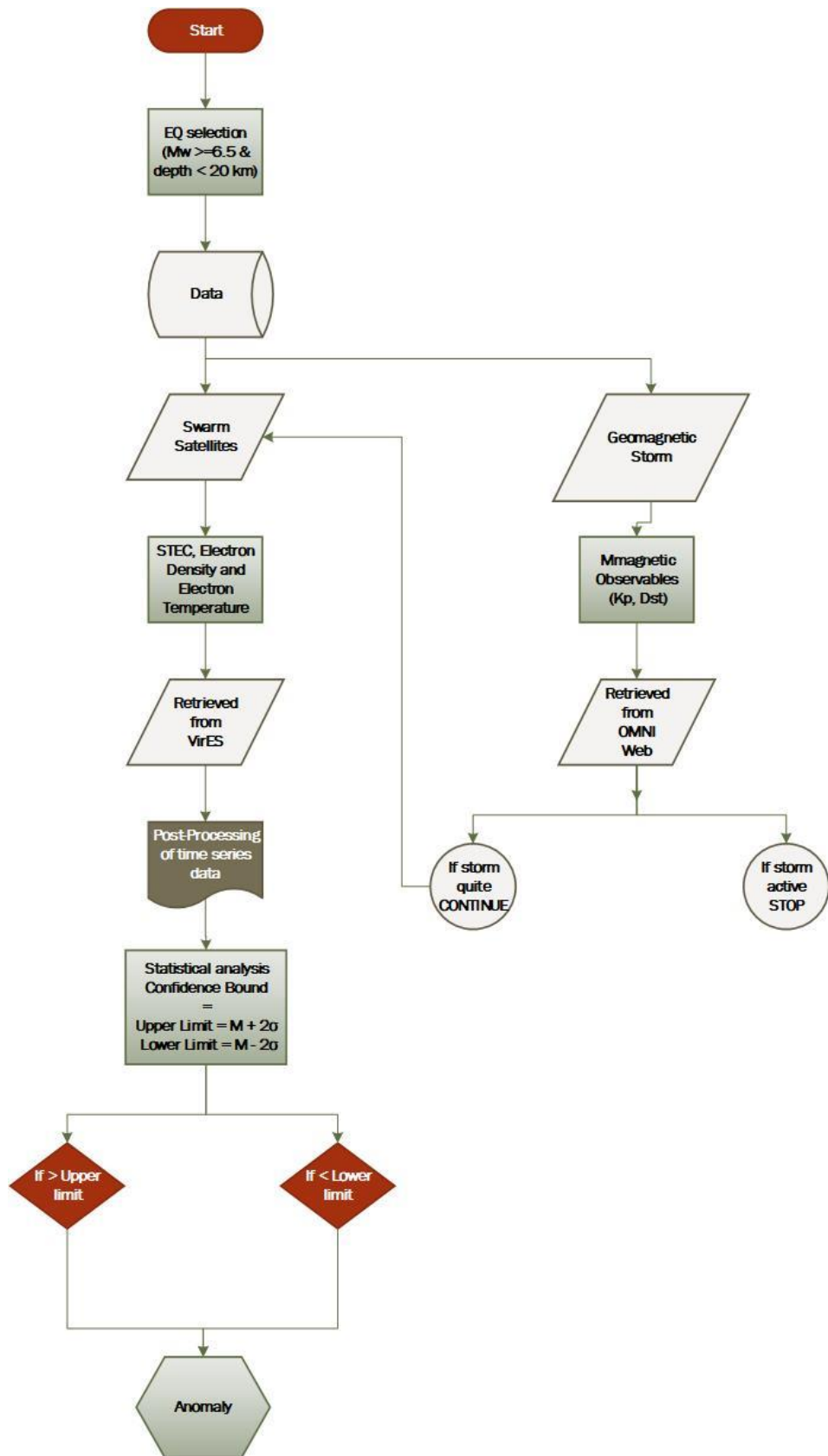
The values of three satellites (Alpha, Bravo and Charlie) are analyzed between the bound of median and standard deviation to identify anomaly in N_e , STEC and ET. These bounds are obtained by the Median (M) and 2σ (σ =standard deviation) of all the values in this study (i.e., 60 days before /after the main shock day). This time period was chosen based on the findings of De Santis et al.[22]. Clear concentrations of electron density and magnetic anomalies were discovered

in his research from almost two months before the main shock until a few days before the main shock. First, Median (Average daily value) and standard deviation (σ) of each ionospheric parameter is taken. Then the upper limit for the bound is obtained by adding Median and double of standard deviation (2σ), whereas lower limit is obtained by subtracting double of standard deviation (2σ) from Median. The formulae for upper and lower limit are given below

$$\textbf{Upper Limit} = \textbf{M} + 2\sigma \quad (4)$$

$$\textbf{Lower Limit} = \textbf{M} - 2\sigma \quad (5)$$

Where median and standard deviation of the data obtained from Swarm satellites is denoted by **M** and σ respectively. The daytime and nighttime atmospheric values of OLR, RH, AT, and ST are also investigated over the epicentral regions estimated by the Dobrovolsky formula. This gives us more evidence to study the variations in the lower atmosphere during the EQ preparation period before and after the main shocks.



3. CHAPTER 3. RESULTS

In this study, the ionospheric perturbations such as N_e , ET and STEC are analyzed from Swarm-three satellites, over the epicentral region of two large magnitude and shallow hypocenter EQs (USA and Turkey events). Similarly, atmospheric indices were investigated in order to find additional evidence for LAIC. Table 1 summarizes all information about anomalies. These variations are studied within the Dobrovolsky region [25] region for the values before/after the main shock to discriminate the seismic and storm responses. Figure 1 depicts the geographical position of both EQs, as well as the Dobrovolsky region for each EQ. Furthermore, the zoomed-in geographical position, Dobrovolsky region, and route of Swarm-three EQ satellites that happened in Challis (USA) and Doganyol (Turkey)EQ are also shown. Furthermore, the geomagnetic activity is depicted in Fig. 2a and Fig. 3a for before and after the EQs. As a result, it is clear that magnetic activity was very low during these two EQs i.e ($K_p < 3$, $Dst < -50$ nT).

In case of the USA EQ, ionospheric anomalies from the Swarm three satellites are analysed in different ionospheric indices to provide additional evidence. Hence, Figs. 2a-2c depicts the geomagnetic activity conditions along with N_e , ET and STEC from Swarm satellites for M_w 6.5, USA EQ on March 31, 2020. As illustrated in Fig. 2a (Swarm A), prominent positive enhancements in ET on EQ Day and a clear peak in STEC associated with the USA EQ are observed; additionally, the increase in ET is greater than 1000 K from the upper bound of statistical bounds' method, and the STEC from the Swarm-A satellite clearly shows an abnormal variation of 3–4 TECU.

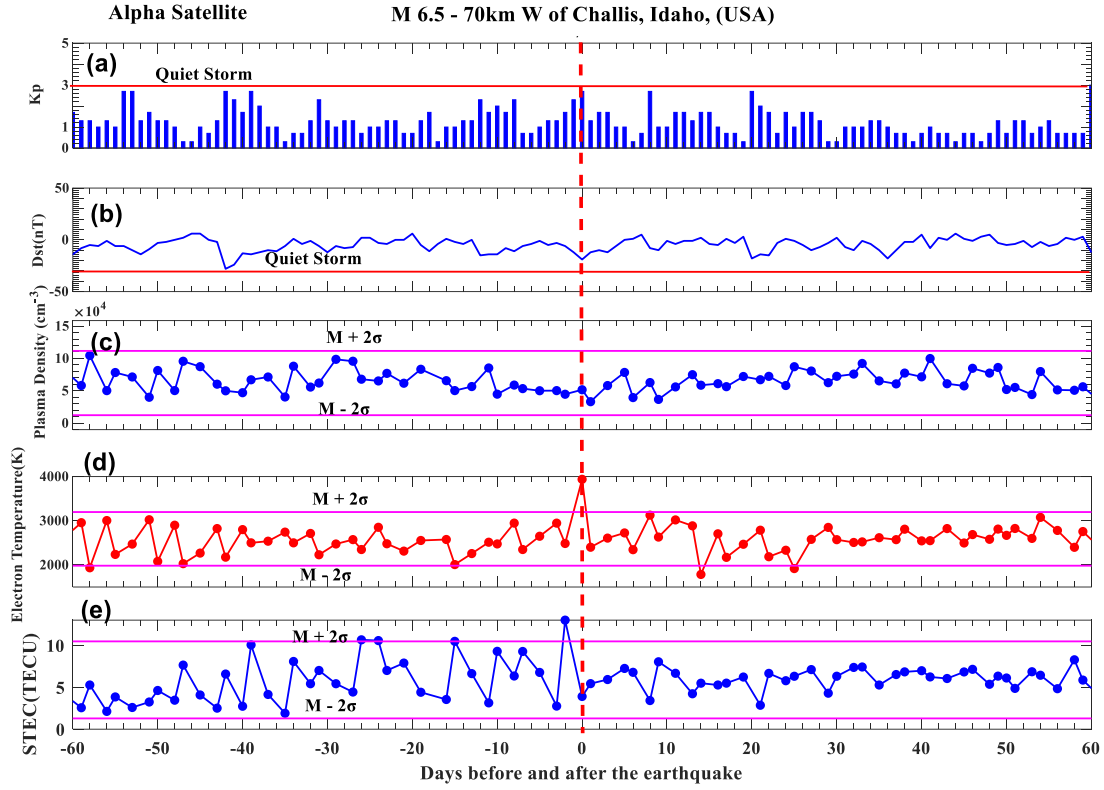


Figure 3.1: (a) and (b) Dst and Kp for (USA). (c) Plasma Density (Ne) (d) Electron Temperature (ET) (e) Slant Total Electron Content (STEC) of Swarm Alpha for M_w 6.5 (USA). The EQ Day is represented by the brown dashed line.

Similarly, the day before the primary shock (March 30, 2020) shows a significant increase in STEC, with Swarm-B values deviating by 3–4 TECU from the rest of the normal distribution (Fig. 2b). These analyses demonstrate that the epicentral regions linked to the USA EQ had abnormal variations in STEC.

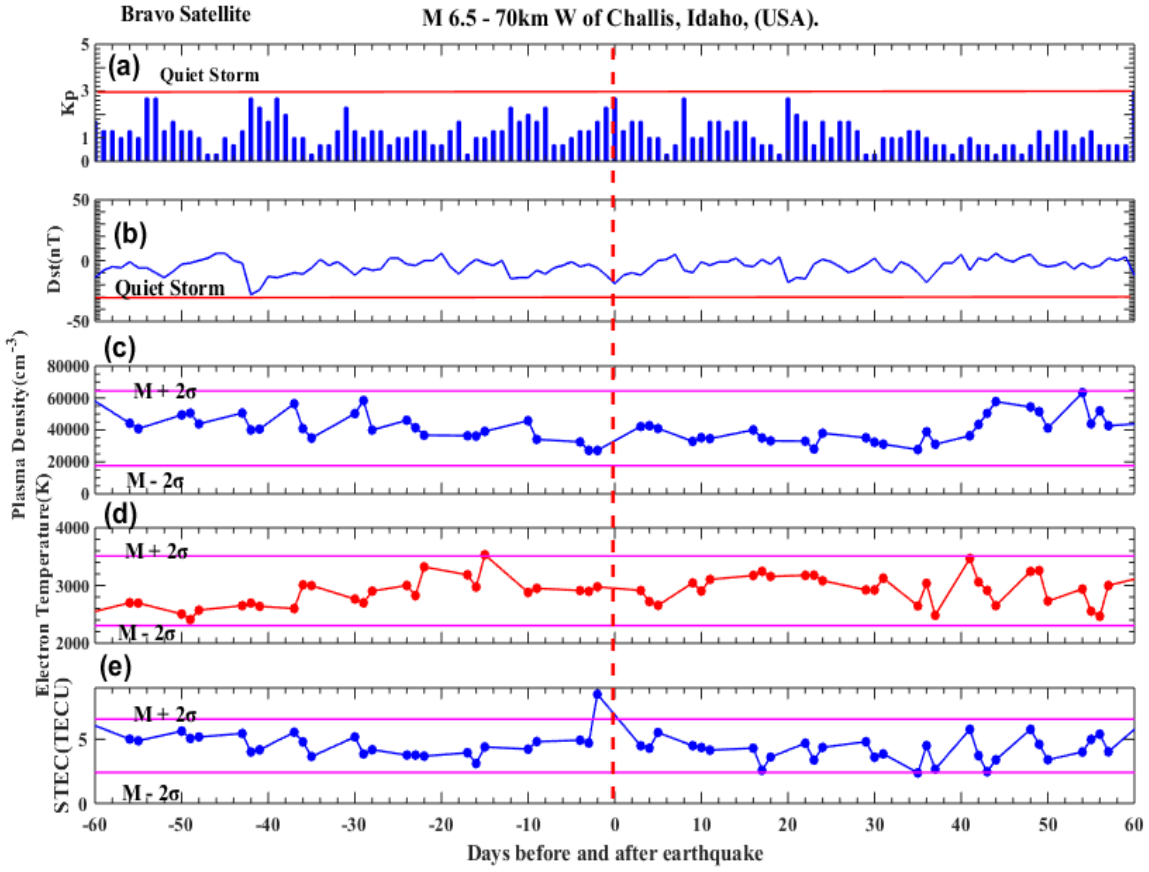


Figure 3.2: (a) and (b) Dst and Kp for (USA). (c) Plasma Density (Ne) (d) Electron Temperature (ET) (e) Slant Total Electron Content (STEC) of Swarm Bravo for M_w 6.5 (USA). The EQ Day is represented by the brown dashed line.

On the other hand, Swarm-C shows no appreciable increase or decline in any of the ionospheric indices in the 60 days prior to/following the USA event (Fig. 2c). As a result of the primary shock in the United States, it is also evident that none of the satellite values show any negative anomalies. The fact that every abnormal pre-seismic anomaly is positive and has a significant value higher than the upper confidence bound suggests that EQ energy was released during the period of preparation. It is also evident that Swarm-A and -B anomalies can be seen during times of quite geomagnetic activity ($K_p < 3$, $Dst < -50\text{nT}$) before the USA EQ.

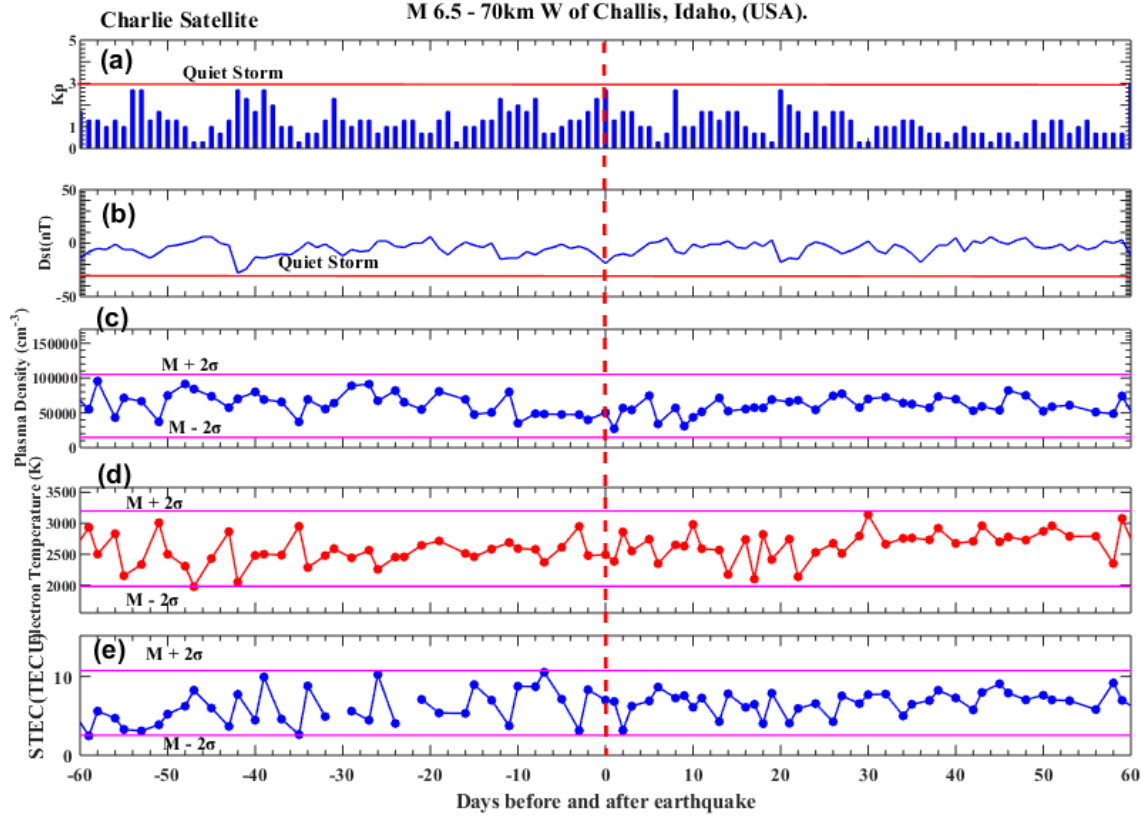


Figure 3.3: (a) and (b) The Dst and Kp for (USA). (c) Plasma Density (Ne) (d) Electron Temperature (ET) (e) Slant Total Electron Content (STEC) of Swarm Charlie for M_w 6.5 (USA). The EQ Day is represented by the brown dashed line.

We also investigate the variation in ionospheric indices (Ne, ET, and STEC) after and before the EQ event of Turkey having M_w 6.7,, which occurs on January 24, 2020. There are no discernible pre-seismic perturbations in any of the three Swarm satellite-based ionospheric indices prior to the main shock. However, prominent peaks in Ne, ET, and STEC are observed within 30 days of the main shock as post-seismic responses. Figure 3a depicts the quiet geomagnetic activity the day before and after the main shock, and it appears that ionospheric attributes would be a clear EQ response. Within 30 days of the main shock, the variations in Ne and ET stand out from the rest of the normal distribution (Fig. 3a). The anomalous perturbations were clearly matched with

the analysis of De Santis et al.[22]; throughout a period of more than a month, they reported seismo ionospheric anomalies.

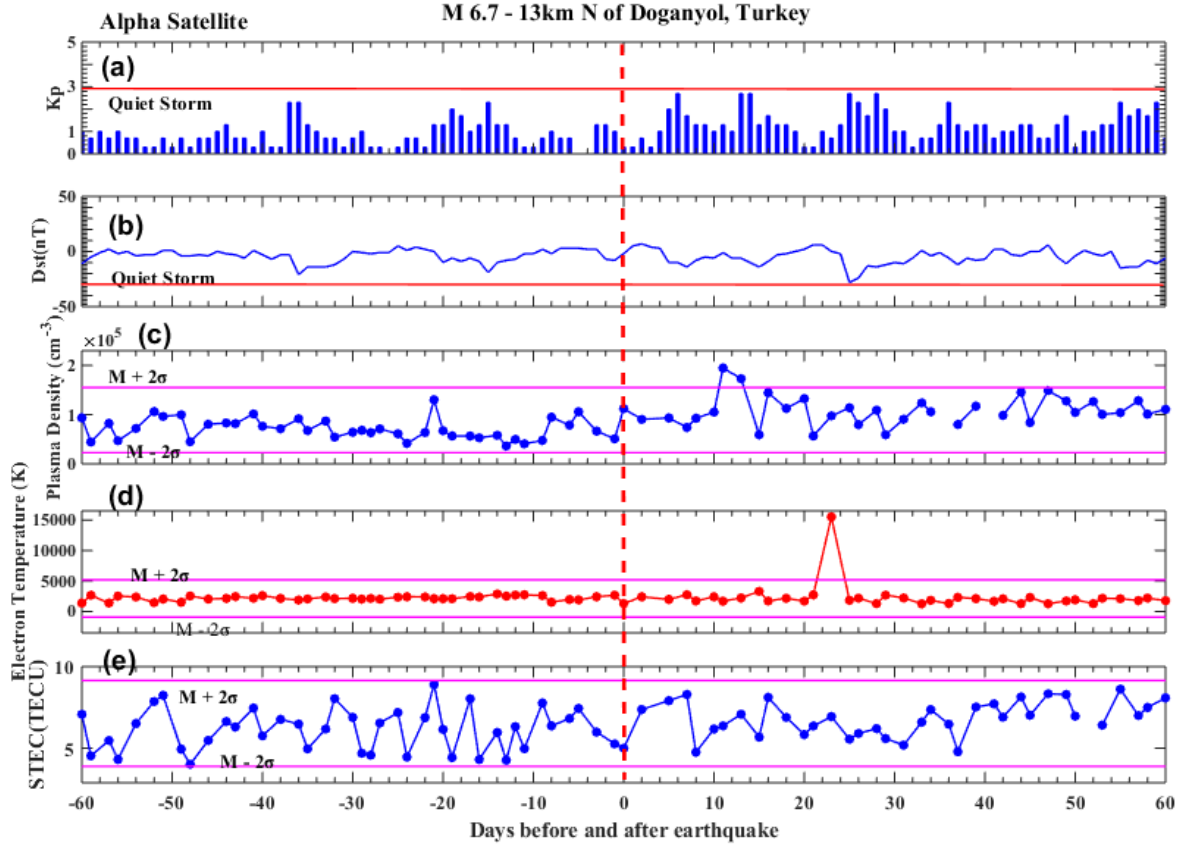


Figure 3.4: (a) and (b) Dst and Kp for (Turkey). (c) Plasma Density (Ne) (d) Electron Temperature (ET) (e) Slant Total Electron Content (STEC) of Swarm Alpha for $M_w 6.7$ (Turkey). The EQ Day is represented by the brown dashed line.

Similarly, in the analysis of the Swarm-B (Fig. 3b) and Swarm-C (Fig. 3c) satellite measurements, abnormal variations in Ne and STEC were found. One month after the primary shock, both satellites as well as the Swarm-A satellite displayed abnormal behavior. Like the USA EQ, all these disturbances are constructive positive changes. This demonstrates the association between positive anomalies and EQ one month before/after the primary shock.

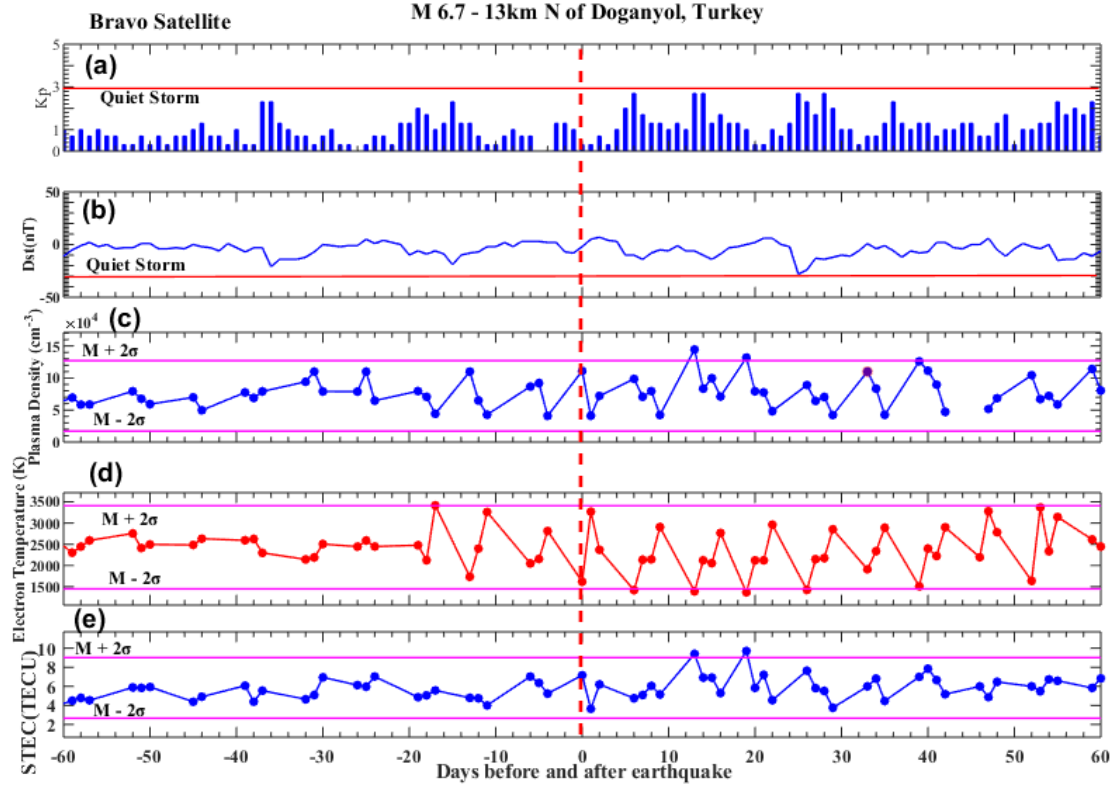


Figure 3.5: (a) and (b) Dst and Kp for (Turkey). (c) Plasma Density (Ne) (d) Electron Temperature (ET) (e) Slant Total Electron Content (STEC) of Swarm Bravo for M_w 6.7 (Turkey). The EQ Day is represented by the brown dashed line.

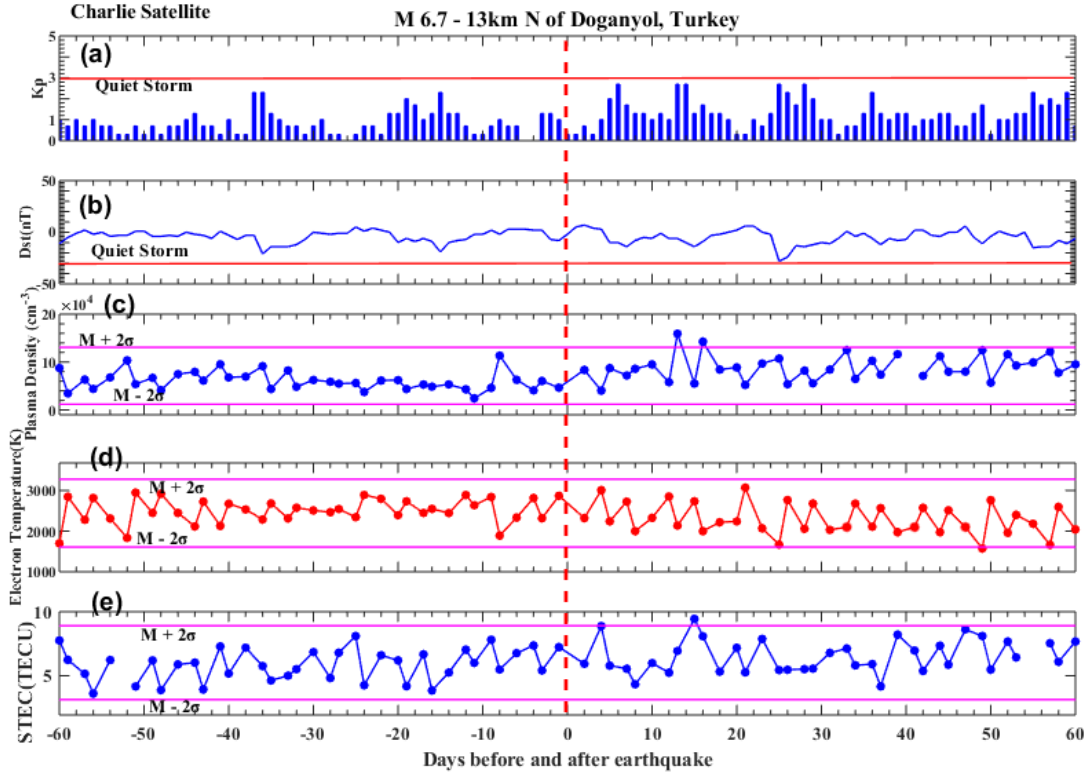


Figure 3.6: **(a)** and **(b)** Dst and Kp for (Turkey). **(c)** Plasma Density (N_e) **(d)** Electron Temperature (ET) **(e)** Slant Total Electron Content (STEC) of Swarm Charlie for M_w 6.7 (Turkey). The EQ Day is represented by the brown dashed line.

Furthermore, daily analysis of N_e , STEC, and ET from the Swarm three satellites reveals distinct variations on the EQ Day of the USA event (Fig. 4) and notable post-seismic responses after the Turkey EQ (Fig. 5). As pre-seismic EQ perturbations, Figure 4 shows more distinct variations of all ionospheric indices from the rest of the observations prior to the main shock. Anomalies in the daily values of the Swarm satellites after Turkey EQ are also clearly visible in Figure 5. The existence of pre- and post-seismic EQ perturbations in the ionosphere is confirmed by all of these variations, but further research is needed to fully comprehend the LAIC coupling above the epicentre.

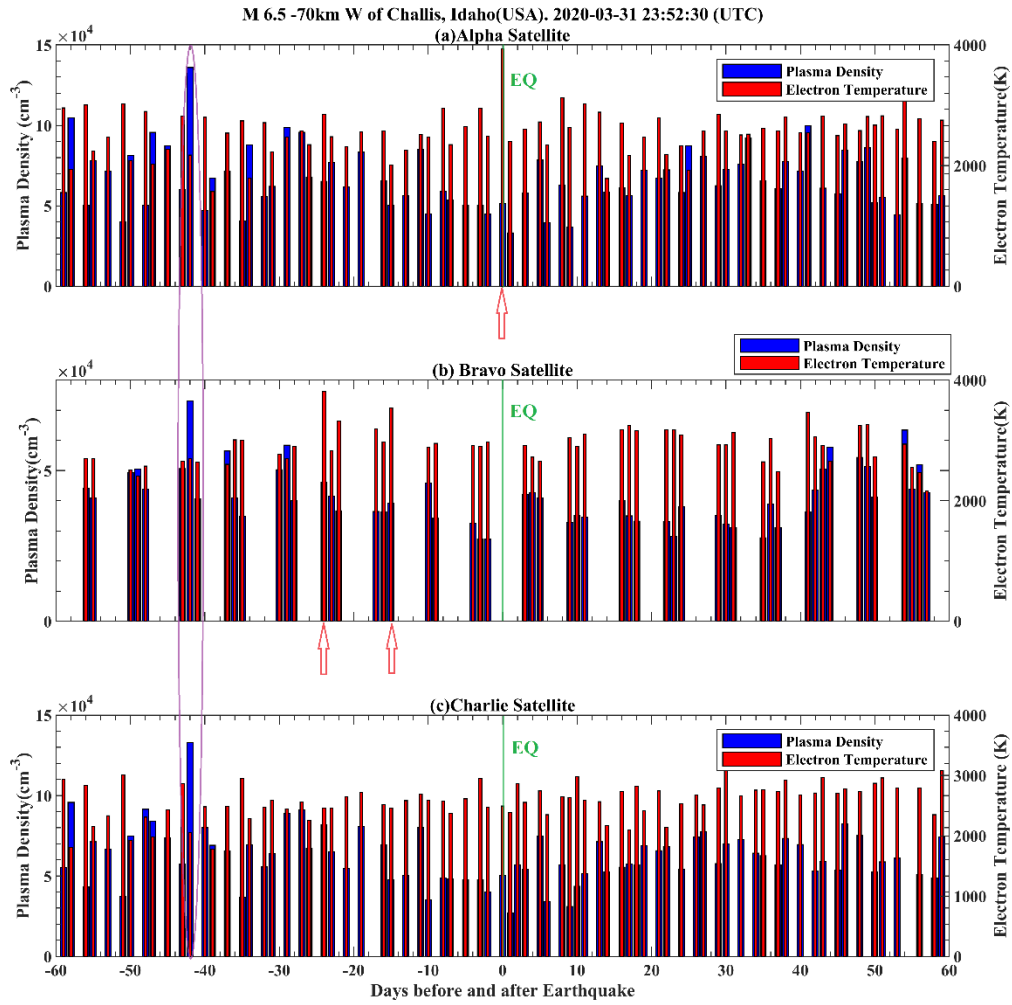


Figure 3.7: Comparison of anomalies from Mw 6.5 (USA) EQ's Swarm A-C. Anomalies in electron temperature (ET) are depicted by red arrows, while anomalies in plasma density (Ne) are indicated by ellipses. EQ Day is depicted by the green line.

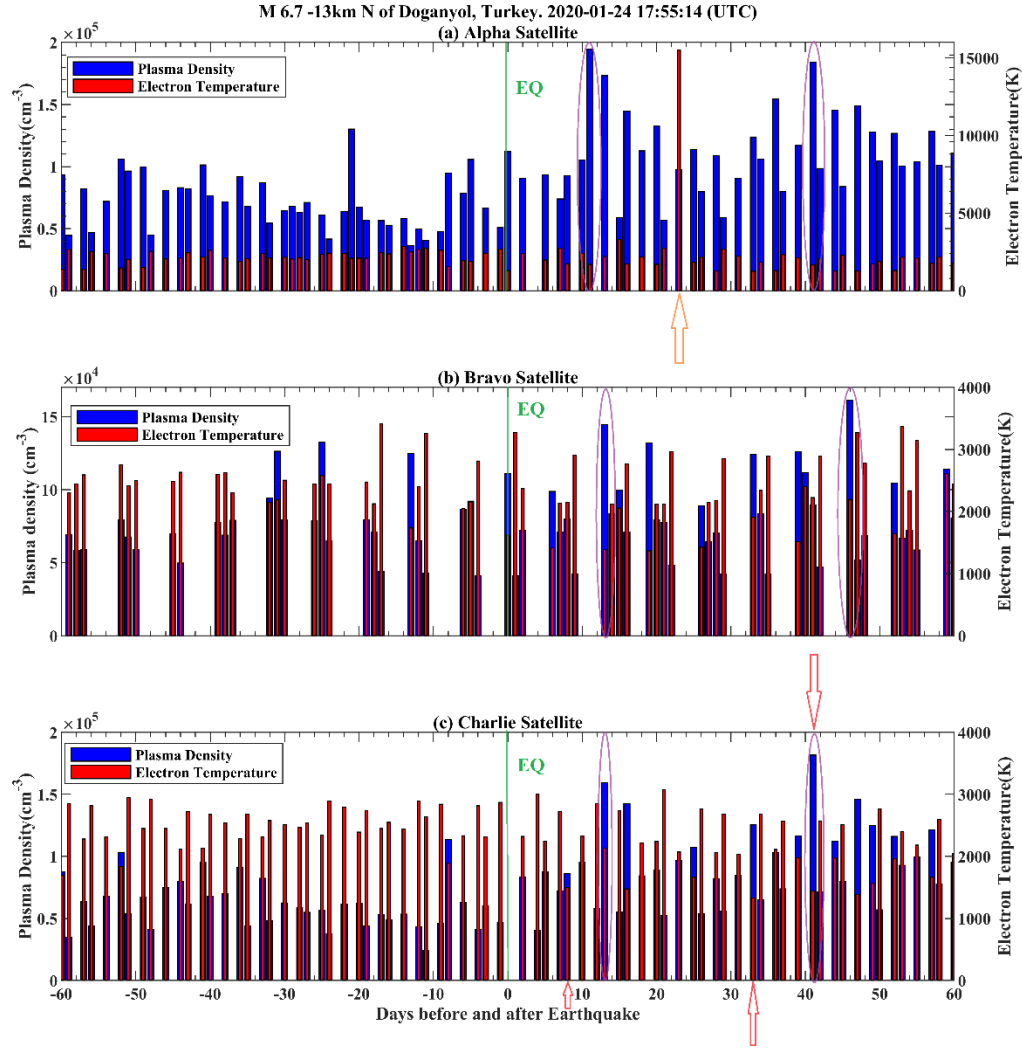


Figure 3.8: Comparison of anomalies from Mw 6.7 (Turkey) EQ's Swarm A-C. Anomalies in electron temperature (ET) are depicted by red arrows, while anomalies in plasma density (Ne) are indicated by ellipses. EQ Day is depicted by the green line.

In order to determine changes in air ionization close to the epicentral atmosphere, averaged values of atmospheric indices for daytime and nighttime for both EQs are also investigated. Prior to the USA EQ, there was a significant variation in the analysis of daytime and nighttime RH as well as daytime ST (Fig. 6).

M 6.5 - 70km W of Challis, Idaho (USA).

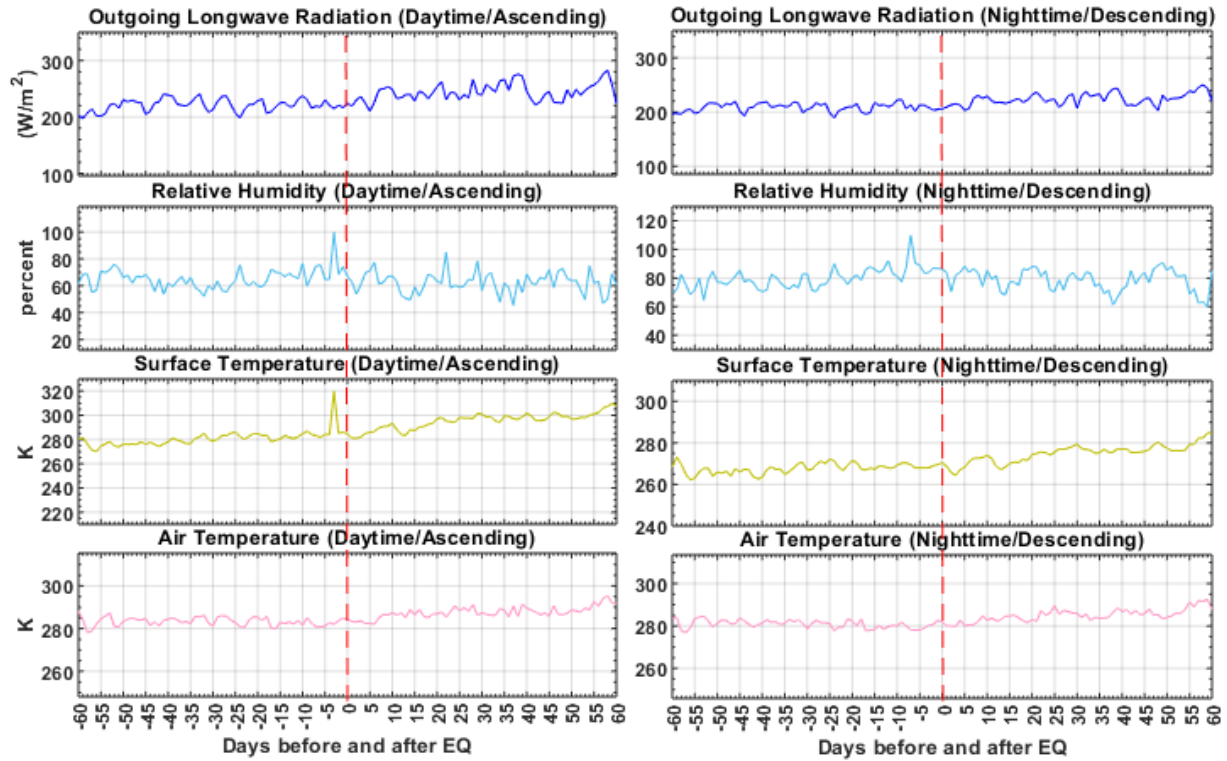


Figure 3.9: Monitoring of atmospheric indices for the USA EQ, where the main shock day is indicated by the red dashed line.

After the Turkey EQ, daytime OLR, ST, and AT show variations for about 15-20 days after the primary shock (Fig. 7).

M 6.7 - 13km N of Doganyol, Turkey.

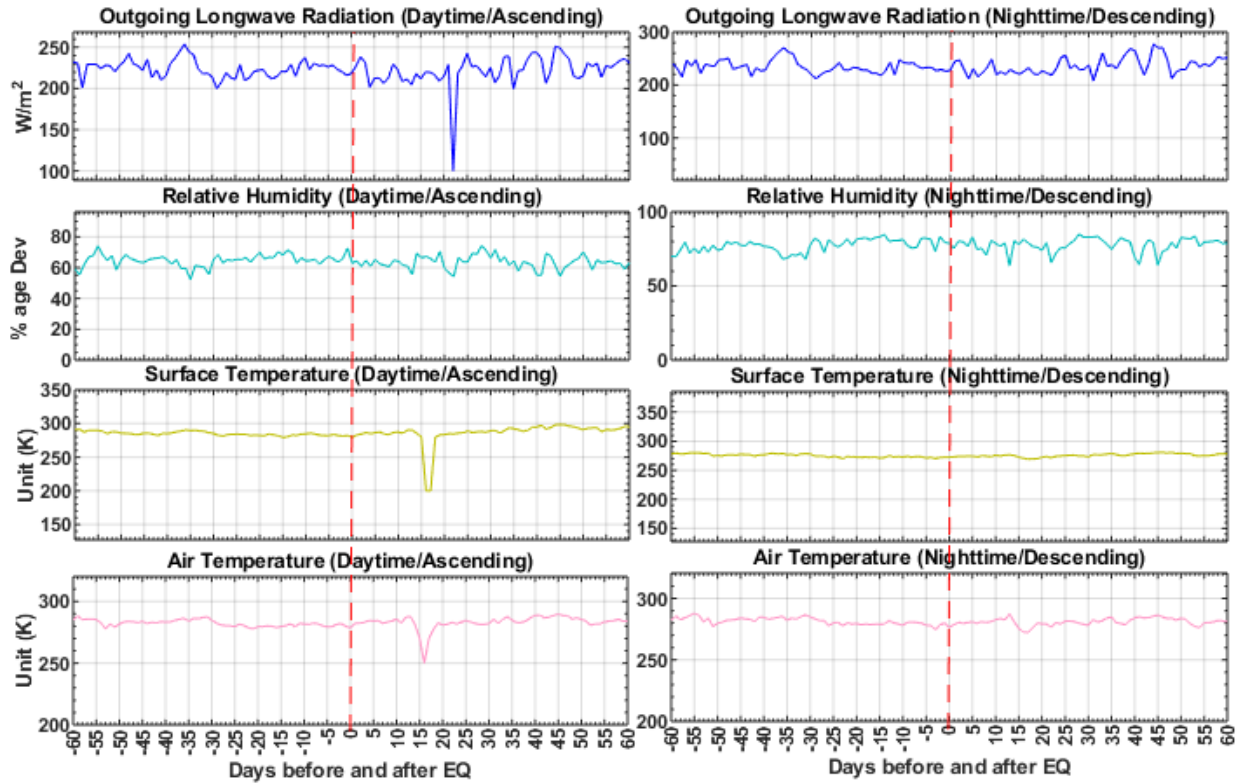


Figure 3.10: Monitoring of atmospheric indices for the Turkey EQ, where the main shock day is indicated by the red dashed line.

The significant peaks of each of these variations coincide with the time period of the Swarm satellite observations, making them stand out from the rest of the data (Table 1). Anomaly patterns in the atmospheric indices have synchronized and co-located with ionospheric perturbations. LAIC coupling can be aided, though, by additional analyses of atmosphere and ionospheric index data from various satellites for various EQs.

4. CHAPTER 4: DISCUSSION

In order to investigate LAIC, an integrated analysis of ionospheric perturbations has been performed such as Ne, ET and STEC for M_w 6.5 USA on March 31, 2020 and M_w 6.7 Turkey of January 24, 2020. These occasions were selected while considering certain thresholds, like shallow depth ($< 20\text{km}$) and $M_w \leq 6.5$. We investigated the various EQs on quiet storm days to quantify the EQ-induced perturbations in order to distinguish between variations of geomagnetic storm and EQs. Before the USA event, there were perceptible changes in the form of earthquakes, and after the main earthquake in Turkey, there were positive post-earthquake changes. This analysis further shows that all the anomalies occur during no storm activity ($K_p < 3$, $Dst < -30\text{nT}$). And all the anomalies occur 2-3 days before the main shock and one-month after the EQ. Additionally, it demonstrates that the EQ energy begins to release more intensely 2-3 days prior to the primary shock and continues to do so for a full month afterwards.

We use Swarm multiple analysis with the three satellites to determine that the two EQs have different ionospheric characteristics. For instance, a month after the main shock of the Turkey EQ, intense ionospheric positive peaks can be seen. Prominent positive enhancements can be seen in Ne, ET, and STEC within 2-3 days prior to the USA EQ. Swarm-C has no variations connected to the USA EQ, which is also noteworthy. However, all other satellites based data have perturbation with both the EQ (i.e., USA and Turkey EQ). The daily ionospheric indices' mutual analysis (Fig. 4) further supports the positive changes that occurred before the main shock and within a month of the Turkish EQ (Fig. 5). Additionally, STEC exhibits greater perturbation in all satellite data related to the two EQ, followed by a sizable change in Ne. It is possible that EQ energy above the epicentre has no effect on ET because there are no appreciable ionospheric variations as a result of EQ.

To determine the capability of ionospheric precursory measurements, the EQ-induced ionospheric anomalies are statistically analysed for global events using long-term data of ground and space observations [16], [18]. Long-term observations are also used to investigate the ionospheric anomalies with Swarm three satellites for various EQ, and it is also concluded that measurements like these can help with EQ forecasting in the future [22]. Numerous studies also discuss the pre- and post-seismic effects of coupling the lithosphere, atmosphere, and ionosphere for various EQs [5]. Additionally [26] stated that during the EQ preparation phase, the tical movement of energy related to EQ to the atmosphere and eventually to the ionosphere was a factor of electrodynamics over the epicentre. Moreover, they also explained that electric field is involved between Ionosphere and the surface of Earth during the seismic phase. Also, at different altitude the change in electric field causes the anomalous cloud over the epicenter. Sorokin et al. [26] showed that $\sim 10\text{m/V}$ electric field on lithosphere can attain the value of $\sim 100\text{mV/m}$ in ionosphere. Similarly in our analysis the perturbations can be seen in Ne, STEC & ET before and after the two EQs from Swarm three satellite ionospheric indices.

The possible precursors of EQs are investigated in atmosphere and ionosphere by utilizing different remote sensing procedures over the epicentral regions for Chile [14], [16], New Zealand [27], Pakistan [28], Turkey (Shahzad et al., 2021) and Japan [29]. Similarly, [3] found evidence of spatial and temporal anomalies in ionospheric data within 5 to 2 days before the main shock of $M_w 7.1$ EQ in Turkey. The existence of EQ precursors in the atmosphere and ionosphere is mentioned in all these reports, but no conclusive results are reached. The aforementioned reports, on the other hand, encourage further work on the hunt for this type of seismic precursors using a larger, better-equipped cluster of atmospheric and ionospheric satellites as well as ground measurements. Numerous ionospheric and atmospheric indices show significant improvement in

this study as well, but more research is required to fully understand the phenomenon. The coupling hypothesis can undoubtedly be improved upon to understand the EQ-related perturbations.

5. CHAPTER 5: CONCLUSIONS

In order to add a new analysis to the LAIC hypothesis, the atmospheric and ionospheric perturbations related to EQs of $M_w \geq 6.5$ in the USA and Turkey using data from Swarm three satellites and Giovanni are analyzed. The findings are described below.

- The geomagnetic conditions such as Dst and Kp within 2 months before and after (window of anomaly) the EQs of (USA) and (Turkey) are quite and Ne, ET and STEC from swarm three satellites show significant perturbations within 0-3 days before the main shock of the M_w 6.5 (USA) EQ. This result is in accordance with the analysis done by Marchetti et al. (2019), in which the reported anomalous window is 30 days before the EQ which occurred in Indonesia (M_w 7.5) by using data from Swarm satellite. On the other hand, M_w 6.7 (Turkey) mostly shows the positive anomalies after the main shock as post-seismic perturbations.
- The EQ anomalies for M_w 6.5 (USA) are present within 2-3 days before the main shock. Data from the Swarm-A and Swarm-B satellites show a significant anomaly, but Swarm-C shows no precursory values. All Swarm satellites for Turkey EQ show anomalies in one or more parameters.
- Additionally, the atmospheric anomalies exhibit sharp positive values prior to the USA EQ and after the M_w 6.7 (Turkey) EQ. This anomalous atmospheric window is in accordance with the previous analysis of Swarm satellites, which provide more evidence to the LAIC coupling.
- The analysis presented supports a deeper comprehension of the physical mechanisms underlying the global preparation for medium-sized to large earthquakes and show the

value of a satellite constellation in monitoring ionospheric and atmospheric seismic anomalies. Lastly, this kind of analysis and its integration with other multi-satellite observations of the atmosphere, ionosphere, and lithospheric data may one day allow us to produce accurate EQ forecasts, preventing loss of life, structural destruction, and economic harm in several nations around the world.

6. CHAPTER 6. REFERENCES

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